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Image-based Barcode Reading Module

User Setup Manual

Version number: v2.0.14

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1 brief statement

This manual mainly provides various function setting commands for barcode reading devices. By familiarizing with the introduction of various function commands in this manual, users can change the function parameters of the barcode reader by scanning the setup code or sending setup commands through the interface. The product has already provided the parameter configurations suitable for most of the usual application functions when it is shipped from the factory, and in most cases, the user can put it into use without making adjustments, and the factory default is the UART serial port output, and the default baud rate is 9600.

Attention:

（1） All setup parameters with * are system defaults;

（2） The CRC checksums included in the commands in this document are CRC16_XMODEM, polynomial x 16+x12+x5+1 (0x1021), data inverted to MSB First, and data formatted as a string format configuration. For example, a command is: $xxxxxxx-yyyyy, the yyyyy 4 bits are the check digit, and the $xxxxxxx-8 bits are the data for calculating the checksum.

2 System settings

2.1 Setting Code Switch

This function is for the user whether can scan the setting code to set the code scanning device. If you turn off the setup code, you can't scan the setup code to configure the parameters, the setup code is always regarded as ordinary barcode; if you turn on the setup code, you can scan the setup code to configure the parameters. The default is to turn on the setup code.

The off setup code command is: $010000-525C **The on setup code command** is: $010001-616D

Note: These two commands can only be set by sending commands through interfaces such as UART.

2.2 Setting code data output switch

This function is whether or not the user outputs the scanned setup code data via the UART interface.

$010100-24E8 (*No setup code data output)		$010101-17D9 (Outputs setup code data)

2.3 parameter management

2.3.1 Restore Factory Settings

Restore all parameters to the factory configuration, the device will reboot automatically after this operation.

$010200-BF34

(factory settings restored)

2.3.2 Saving and restoring user settings

Save User Settings is to save the current settings as user settings;

Restore User Settings is to restore the user's saved settings, if no user settings have been saved, they will be restored to the program's default settings (parameters marked with * inside each setting).

$010207-26A3 (Save user settings)		$010208-369D (restoring user settings)

2.4 Equipment Information

2.4.1 Program version information

$010300-C980

(Firmware version information can be read out)

2.4.2 Device Unique Identifier

Upon successful reading, the device returns a 12-byte string to the host. This can be used to identify the product.

$0103A0-26B8

Read device unique code (12 characters)

2.5 Customized Product Models

2.5.1 Setting up customized product models

With this function, users can name the product model by themselves, and then it can be read out by the read command. The steps to set up a customized product model number are as follows:

(1) Scan to set up a custom product model to start the setup code as follows:



$010204-73F0 Enter customized product model setting status

(2) Make a non-setting QR code for a customized product model and scan the code, the buzzer will sound three times to indicate successful setup, and a long buzzer will sound once to indicate failed setup. As below:



CUSTOM PRODUCT MODEL TEST

Customized product model test code, after setting in step (1) and then scanning the following code, the customized product model will be set as CUSTOM PRODUCT MODEL TEST, which can be read out by scanning the command code of 2.5.2 Read Customized Product Model or sending commands.

Attention:

a. The total characters of the customized product model cannot exceed 100 bytes, otherwise the setting will fail;
b. You need to restart the setup from step (1) when the setup fails;
c. If you want to modify the setup model after successful setup,

restart the setup directly from step (1), and the original setup model will be cleared;

d. After successful setup, restoring the factory settings also retains the model number it was set to.

2.5.2 Read customized product models

Reads customized product models set by the user without buzzer alerts.



$010205-40C1 (reads custom part number)

2.6 Device reboot

The device will reboot immediately after scanning the setup code or after sending the setup command.



$010500-EE19 (Device reboot)

3 Communication settings

3.1 Interface Settings

The factory default is UART interface output, if you need other interface output, scan the enabling code of the corresponding interface to switch, the setting will not take effect immediately, you need to manually restart the device to take effect, and its scanning to restore the factory settings will not be restored to the set interface, or to maintain the original settings unchanged.

3.1.1 UART Serial Output

Switching the data output interface to UART serial output will not take effect immediately and requires a manual reboot to take effect. The default output configuration of the serial port is: Baud rate - 9600bps, Stop bit - 1 bit, Data bit - 8 bits.
bit, parity bit - none (none), no flow control. The baud rate can be changed according to the settings in subsection 3.1.3.



$020003-C9EF

UART Serial Output

3.1.2 UART Serial Port Job Setup

The default is to keep sending in UART serial port regardless of the interface output mode, receiving will be off in USB virtual serial port output mode and on in other modes, which can be set to use the serial port to transfer data only in UART serial port output mode.
(Note: The default parameter for v4.x.39, v6.0.9 and later is "Only works in UART serial output mode", and the default parameter prior to v4.x.39, v6.0.9 was "Maintains operation regardless of interface output mode". '').



$020E01-51BD

Maintains operation regardless of interface output mode



$020E00-628C

*Only works in UART serial output mode

3.1.3 UART serial port baud rate setting

The baud rate of the serial port is adjustable, the default bit is 9600, which takes effect immediately after setting.

 $020503-75AA (Baud rate 2400)	 $020504-EC3D (Baud rate 4800)
 $020505-DF0C (* baud rate 9600)	 $020506-8A5F (baud rate 19200)
 $020507-B96E (Baud rate 38400)	 $020508-A950 (Baud rate 57600)
 $020509-9A61 (baud rate 115200)	

3.1.4 USB HID Keyboard Output

Data output toggle bit USB HID keyboard interface output, will not take effect immediately, need to manually reboot to take effect.

 $020000-9CBC (USB HID Keyboard Interface)

3.1.5 USB HID Keyboard Output Chinese Setting

USB HID keyboard output in Chinese is currently only available for windows systems, and is not supported on Android and li

Outputs Chinese on nux devices. Chinese output is disabled by default.

 $020A00-A87D *Disable USB HID keyboard Chinese output	 $020A02-CE1F Enable USB HID keyboard Chinese output

3.1.6 USB HID Keyboard Output Interval

The USB HID keyboard is transmitted character by character. This setting allows you to change the speed of the embedded scanner's character output to suit the host's requirements. The default is 10ms interval.

 $020700-CD91 No spacer	 $020701-FEA0 Interval 1MS
 $020703-98C2 Interval 3MS	 $020705-3264 Interval 5MS
 $02070A-F6F9 *Interval 10MS	 $020714-3665 Interval 20MS
 $02071E-0D0D Interval 30MS	 $020732-F2A3 Interval 50MS

3.1.7 USB HID Virtual Keyboard Numeric Keypad Selection

 $020900-6FCB *Alphabetic keypad	 $020901-5CFA Numeric Keypad

3.1.8 USB HID Keyboard Layout

The USB HID keyboard defaults to the USA keyboard output.

 $020C00-4515 *USA	 $020C01-7624 Belgium	 $020C02-2377 Suomi
 $020C03-1046 French	 $020C04-89D1 German	 $020C05-BAE0 Sweden
 $020C06-EFB3 United Kingdom of Great Britain and Northern Ireland	 $020C07-DC82 Norway	 $020C08-CCBC Spanish
 $020C09-FF8D Portugal	 $020C0A-7E7D Turkey Q	 $020C0B-2B2E Japanese
 $020C0C-181F Czech Republic (from 1993)	 $020C0D-8188 Turkey F	

3.1.9 USB Virtual Serial Output

Switching the data output to USB virtual serial port output will not take effect immediately and requires a manual reboot to take effect.



Serial Port

In window system, you need to install the driver of USB virtual serial port (win10 system may install it automatically) and then use the serial port debugging assistant software to open the port for debugging or development. In linux system, there will be a new ttyA CM node.

3.1.10 USB HID POS output

USB HID POS mode, PID is 0xA4AF, VID is 0x0525, packet size is 32 bytes, input is 32 bytes, output is 32 bytes, there will be packet header and packet tail before and after the module sends data.

The data in the packet header is: the number of QR code data bytes + the length of the checksum;

The data of the packet is sent 32 bytes at a time and a checksum is added at the end of the data, which is the value of the sum of all the data of the QR code data;

The packet tail is a packet containing only one byte 0X0D.

Therefore, the data to be sent for a segment with a data length of 80 bytes is.

name (of a thing)	digital	Packet length
Baotou prefecture level city in Inner Mongolia	Number of bytes of QR code data + length of checksums The checksum length is 4 bytes, with the first byte being the lowest bit. Byte 4 is the highest bit	32 bytes
Packet 1	QR code data 1-32 bytes	32 bytes
Packet 2	QR code data 32-64 bytes	32 bytes
Packet 3	QR code data 64-80 bytes + 4 bytes checksum. The checksum length is 4 bytes, with the first byte being the lowest bit. Byte 4 is the highest bit	32 bytes
finish off	A packet containing only one byte, 0X0D.	32 bytes

Please refer to Appendix 5 for details.

(Note: This feature is only supported in FW v4 series versions with v4.x.39 and higher.)



3.2 Host Command Response Mode

The answer mode to return the setup status when sending setup commands using the interface. The default is no answer. **No answer**: No answer data is returned after sending a setup command for successful or failed setup.

Answer mode 1: Answer is fixed data. The answer is $000100-6148 if the command is successfully executed, and $0001FF-A91F if the command is not executed.

Answer Mode 2: The answer is the same as the first 5 characters of the command sent to the device; the 6th character is the same as the first 5 characters of the command sent to the device.

~The 8th character is fixed character "-"; the 8th to 12th characters are CRC16 checksums. The eighth character is fixed character "-"; the eighth to twelfth characters are CRC16 checksums.

Note: (1). Commands of the read class (e.g., read device information) are not answered and reply directly to the read data;

(2). Sending "Enter online upgrade status" will only reply that the command was executed successfully, and it only means that the command has been received.

$020B00-33A1 (*No answer)	$020B01-0090 (Response mode 1)	$020B02-55C3 (Response mode 2)

3.3 Heartbeat Search

3.3.1 Heartbeat Mode 1

The host can send a heartbeat query command to see if the code-scanning module has a heartbeat response, so as to determine whether the code-scanning module is working normally.

Heartbeat Inquiry:
$01036F-D9DF
Heartbeat Response:
$010370-4C10

3.3.2 Heartbeat Mode 2

The following occurs when heartbeat mode 2 is enabled:

1. heartbeat sends 0x82 0x83 every 10 seconds; 2. power up reboot power up



383700-7DC3

*Disable heartbeat

mode 2



$383701-4EF2

Turn on heartbeat

mode 2

4 Reading parameter setting

4.1 recognition mode

In order to apply to different scenes, the code scanning module has various reading modes, and the default is auto-sensing reading mode. The settings are effective immediately.

4.1.1 Auto-sensing mode

In auto-sensing mode, when a barcode enters the field of view of the device or presses the key once, the device will start reading once. When the reading is successful or the reading exceeds the "single reading time", it will stop reading.



$100004-6359 (*Autosensing)

4.1.2 continuous reading mode

In Continuous Reading Mode, the device reading will be turned on continuously and is not affected by the "Single Reading Duration".



$100003-FACE (continuous literacy)

4.1.3 keystroke mode

In the key reading mode, the code scanning will be controlled by key presses. There are 3 control modes. In order to prevent mis-scanning which may lead to failure of code scanning if no key is connected, this mode is recommended to be set by sending setting commands from the applicable interfaces, so as to ensure that the settings can be set through the setting

commands.

（1） **(a) Key once trigger mode**: press the key once to start reading. When the reading is successful or exceeds the "single reading time", the device stops reading;

The order is: **$100000-AF9D**

(2) **Key Hold Mode**: In this mode, the device starts reading when the key is pressed and stops reading when the key is released. The device stops reading when the reading is successful or when the "single reading duration" is exceeded;

The order is: **$100001-9CAC**

(3) **Key Switch Continuous Mode**: In this mode, the key acts as a toggle switch, starting continuous reading when pressed and stopping when pressed again, without being affected by the "single reading duration" during the opening period.

The order is: **$100002-C9FF**

4.1.4 Command Trigger Mode

Command Trigger Mode should be applied when the key is triggered once and the key is held or then in auto-sensing mode, and the command trigger in Command Trigger Mode is read as follows:

Command Trigger 1: Trigger reading after sending a command, when the reading is successful or exceeds the "single reading time" or receives the "command trigger stop", the device stops reading;

The command is: $108000-ADB0

Command Trigger 2: The command is sent to trigger continuous reading, and the device stops reading only when the command "Command Trigger Stop" is sent;

The order is: $108001-9E81

Command Trigger Stop: Used to stop the reading triggered by "Command Trigger 1" and "Command Trigger 2". The command is $108003-F8E3.

4.2 Auto-sensing sensitivity

This setting is only valid when the "reading mode" is "auto-sensing"; the higher the sensitivity, the easier it is to trigger the device to enter the reading state, and the lower the sensitivity, the less likely to trigger the device to enter the reading state, you need to set the sensitivity according to different scenarios, in general, the default settings can be used. In general, the default settings can be used. The default is the highest sensitivity.

 $100100-D929 *Highest auto-sensing sensitivity	 $100101-EA18 High auto-sensing sensitivity	 $100102-BF4B Medium auto-sensing sensitivity
 $100103-8C7A Low auto-sensing sensitivity		 $100104-15ED Minimum auto-sensing sensitivity

4.3 Auto-sensing restart delay setting

This setting is only valid when "Reading Mode" is "Auto Sensing"; it is used to set the delay time for the next auto-sensing activation after the barcode is read; it is recommended that the delay time should not be set too short, otherwise it is easy to produce multiple readings in one scanning. The default is 0.5s.

$101304-52D4 (0.3 seconds)	$101306-34B6 (*0.5 sec.)
$101308-17B9 (0.7 seconds)	$10130C-C31A (1 sec.)

4.4 Auto-sensing key mode

You can select the mode of key trigger reading under auto-sensing. One Time Read Key Mode: Pressing a key forces a one time auto-sensing read per unit of time. Key Hold Key Mode: After pressing the key once or twice in a row, the module will enter the manual mode, in which the manual mode will be activated.

When the module is in auto-sensing mode, the module will not automatically trigger the reading. Pressing the key will trigger the reading and releasing the key will stop the reading immediately. When the module stops reading and is idle for 5 seconds, the module will re-enter auto-sensing mode. (Note: FW v4 series versions are only supported by v4.x.29 or later; FW v6 and FW v1 series versions support this feature).

$101500-B989 (*One time read mode)	$101501-8AB8 (key hold mode)

4.5 Single reading time

This setting is only effective when the "reading mode" is "one time trigger", "key hold", "auto-sensing"; the device will stop reading when the reading time exceeds the "single reading time". This setting is valid only when "Read Mode" is "One Time Trigger", "Key Hold" or "Auto Sense"; when the reading time exceeds "Single Reading Duration", the device will stop reading. The default value is 6 seconds.

$100200-42F5 (unlimited length)	$100203-17A6 (3 seconds)	$100206-E853 (*6 seconds)
$100209-F86D (9 seconds)		$10020F-E00A (15 seconds)

4.6 Autosensing Mode Read Completion Mode

When the auto-sensing mode recognizes the data, whether it enters the code-sweeping state again, exits the code-sweeping state immediately, or ends the auto-sensing mode 3: After the code-sweeping is successful, the fill-in light will flash for a while, and then continue to keep the triggering state.

(Note: For FW v4 series version, the default is "Exit Trigger State Immediately" and is only supported by v4.x.27 and higher; for FW v6 series version, the default is "Auto Sense End Mode 3" and is only supported by v6.09 and higher). v6 series version, default "Auto Sense

End Mode 3", and only supported by v6.09 and higher).

$101400-CF3D * Immediate exit from the triggered state	$101401-FC0C Continue to hold the trigger state	$101402-A95F Auto-sensing end mode 3

4.7 Manual query of motion detection status

Use the query command: $380000-C23C Return without physical movement: $380010-F50C Return with physical movement: $380011-C63D

$380000-C23C

4.8 Automatic sending of motion detection status

This function is only effective when the reading mode is auto-sensing mode, after enabling, if physical movement occurs, it will send $380211-2B55 data to the host computer once within 3s.

$380200-2F54 *Not enabled

$380201-1C65 Enable

4.9 Barcode reading interval

This setting is only effective when the "Reading Mode" is "Auto Sensing", "Continuous Reading" or "On/Off Continuous"; it is used to adjust the interval time between reading barcodes to prevent the same barcode from being read several times due to too fast reading speed. This setting is only effective when "Reading Mode" is "Auto Sensing", "Continuous Reading" or "Switch Continuous"; it is used to adjust the interval time of reading barcode to prevent reading the same barcode several times due to the reading speed is too fast.

Sweep interval time: Interval between certain time to solve the code once. The default is 0.5s.

No duplicate output within x seconds of neighboring barcodes: The same barcode must be out of the read zone for more than x seconds before it will be read again.

(Note: "Adjacent barcode repeated within 0.3 seconds is not output" and "Adjacent barcode not repeated within 0.5 seconds is output" only)

v4.x.39 and higher).

 $100300-3441 (Sweeps without intervals)	 $100305-CBB4 (* Scanning interval 0.5 seconds)
 $10030A-0F29 (1 second between sweeps)	 $100314-CFB5 (2 seconds between sweeps)
 $1003FB-30D2 (adjacent barcodes are not duplicated within 0.3 seconds)	 $1003FD-9A74 (adjacent barcodes are not repeated within 0.5 seconds)
 $1003F0-5EE9 (Adjacent barcodes are not duplicated within 1 second)	 $1003F2-388B (Adjacent barcodes are not duplicated for 2 seconds)
 $1003F3-0BBA (Adjacent barcodes are not duplicated within 3 seconds)	 $1003FF-FC16 (adjacent barcodes must be different)

5 Lighting and aiming

5.1 illuminated lamp

5.1.1 Illuminated mode

The illuminator provides auxiliary illumination for photo reading, with the light beam shining on the reading target to improve reading performance and adaptability in low-light environments. Users can set it to one of the following states according to the application environment (Default is to light up when recognized):

Always on: the lighting is continuously illuminated after the device is turned on; it is recommended to set it to always on when the reading environment is dark or completely devoid of light.

No illumination: the illumination lamp does not light up under any circumstances; it is recommended to set it to no illumination when the reading object is a screen. **On when reading**: the light is on when reading and off at other times.

$140000-A93C (Changliang)	$140001-9A0D (no lighting)	$140002-CF5E (*Lights up when read)

5.1.2 Illuminated mode 2

This mode command is only valid for the current time, after power off and reboot, it returns to the state before setting.

$140006-039A (always on, not saved)	$140007-30AB (no illumination not saved)

5.1.3 Lighting control method

The lighting control mode supports PWM control or level control, when it is PWM mode, you can set the lighting brightness, when it is level mode, you can set whether the lighting is low level or high level. The default is PWM control, you can use the following setting code to set the brightness of the light.



$382500-3AFA

level mode



$382501-09CB

*PWM mode

5.1.4 Lighting operating level setting

Note: You need to set the lighting to level mode first, the setting takes effect immediately, and restoring the factory setting will restore it to the default value.



$382600-A126 Illuminated low level on



$382601-9217 *Lighting lamp on high

5.1.5 Lighting Brightness Setting

Brightness can only be set when the lighting control mode is PWM mode. (Note: For FW v4 series version, "Brightness Darkest" and "Brightness Very Dark" settings are only supported by v4.x.39 and later versions; FW v6 series versions all support this function).



$150905-E01E Brightest Brightness



$150900-1FEB Brightness Dim



$150902-7989 Brightness



$150904-D32F Brightness is dim.



$150901-2CDA *In Brightness



$150903-4AB8 Brightest Brightness

5.1.6 Automatic fill light in dark places

When this function is turned on, in auto-sensing mode, the module can detect the ambient brightness, and when the ambient brightness is too low, it can automatically turn on the fill light at a lower fill light brightness. (Note: FW v4 series version, and v4.x.24 and higher version support this function, FW v6 series version, and v6.x.14 and higher version support this function, FW v1 series version does not support this function for the time being).

$140300-32E0 *Disable auto fill light		$140301-01D1 Turn on the auto fill light

5.1.7 Auto Fill Light Threshold

This function is detailed in "5.1.5 Auto fill light".
This function sets the brightness at which the fill light will automatically light up, (Note: Only FW v4 series version and
This feature is supported in v4.x.24 and later).

$140200-4454 *Bright at darkest br		$140201-7765 Bright when dark
$140202-2236 Brightness Medium Time Bright	$140203-1107 Bright when high	$140204-8890 Brightness Bright at Brightest

5.1.8 Auto fill light turn on interval

This function is detailed in "5.1.5 Auto fill light".

This function sets how long the fill light will come on after the ambient brightness falls below the threshold, and in the time before it comes on, once the ambient brightness exceeds the threshold, it will be re-timed, which defaults to 5 seconds. (Note: This function is only available for the FW v4 series version, and v4.x.39 and later versions.)

$140400-63CD interval-free	$140405-9C38 *5 seconds interval	$14043C-6797 Interval 60 seconds
$140478-6FF4 Interval 120 seconds	$1404B4-1961 Interval 180 seconds	$1404FF-AB9A Interval 255 seconds

5.2 sighting light

5.2.1 Aiming Light Mode

The aiming beam helps the user to find the optimal reading distance when shooting and reading, which is only available for modules with aiming light. Users can choose any of the following modes according to the application environment (the default is aiming while reading):

Always on: Continuously projects an aiming beam.
No aiming: the aiming beam is extinguished in all cases. **Aiming while reading**: the aiming beam is projected only while reading.

$140100-DF88 (Chang Liang)	$140101-ECB9 (no targeting)	$140102-B9EA (*Aiming while reading)

5.2.2 Aiming light blinking

Aiming Light On Whether or not the light blinks when the light is on, by default it is on without blinking.



$140104-134C *Non-flashing



$140105-207D Flashing

6 Output tips

6.1 Power Up Alert

If the power-up prompt is turned on, the buzzer and light will signal when the device boots up successfully, which is turned on by default.

 $150000-EC9C (turn off power-up alert)	 $150001-DFAD (*Enables power-up alert)

6.2 reading cue

If Read Beep is turned on, the buzzer will beep once after each successful decode and read. This is enabled by default.

 $150100-9A28 (turn off read alert tone)	 $150101-A919 (*turns on read alert tone)

6.3 Setting the tone

If Setup Beep is turned on, the buzzer will sound every time a parameter is set, depending on the result of the setup (3 beeps for a successful setup, 1 long beep for a failed setup). This is enabled by default.

 $150A00-D85D (turn setup beep off)	 $150A01-EB6C (*Turn on setup tone)

6.4 LED indicator

6.4.1 LED indicator on/off

If LED indication is turned on, the LED will blink once after each successful reading. The default is on.

 $150200-01F4 (off LED indication)	 $150201-32C5 (*On LED indication)

6.4.2 Setting the reading result indicator operating level

Note: Settings take effect immediately, restoring the factory settings will restore them to the default values.

 $382700-D792 *Read Result Indicator Low Power	 $382701-E4A3 Read result indicator high

6.5 Lighting control signal output

If enabled, the original LED indicator function will be disabled automatically, and the illumination of the scanning module will be turned off, and the illumination control signal will be output from the LED indicator control IO port, and a high level indicates that the illumination is on, and a low level indicates that the illumination is off. Default disable.

 $150600-CB05 (*prohibited)	 $150601-F834 (enable)

6.6 Power-Up Command Prompt

If enabled, the command " $150600-CB05" will be sent to the host after the device startup is complete, indicating that the device is ready.

$150500-50D9 (*prohibited)	$150501-63E8 (enable)

6.7 Buzzer reads the duration of the beep

When the scanning code is successfully read once, the buzzer will beep once, the default is 60ms.

$150303-2213 (beep duration 30ms)	$150306-DDE6 (Beep duration *60ms)
$15030C-2A4A (Beep Duration 120ms)	$150314-8CB4 (200ms chirp duration)

6.8 Buzzer signal output mode

Set the buzzer control signal, the default is PWM signal to control the buzzer output. The setting will not take effect immediately, you need to restart manually after setting to take effect.

$150400-266D (level output)	$150401-155C (*PWM output)

6.9 Buzzer Frequency

This function can only be set when the buzzer control signal is in PWM mode (section 6.8 can set the control signal output). Set the appropriate control frequency according to the buzzer frequency, the default is 2.7KHZ.

 $150705-4244 (Buzzer frequency 2KHz)	 $150706-1717 (buzzer frequency 2.4KHz)
 $150707-2426 (*Buzzer frequency 2.7KHz)	 $150708-3418 (buzzer frequency 4KHz)

6.10 Buzzer Loudness Adjustment

This function can only be set when the buzzer control signal is in PWM mode (Section 6.8 allows you to set the control signal output), and adjusts the loudness of the buzzer. The default is loudest.

 $150801-5A6E (*high loudness level)	 $150802-0F3D (medium loudness level)
 $150803-3C0C (low loudness level)	 $150804-A59B (lowest loudness level)

6.11 Buzzer operating level

The setting is valid only when the buzzer signal output mode is "Level Output".

 $382800-037C Buzzer low level	 $382801-304D *Buzzer high level

7 Decoded Data Editing

Description:

(1) Data format:

prefix (optional, up to 20 bytes)	AIM ID	Barcode Type Name	user data	suffix (optional, up to 20 bytes)	CRC	closing character
					Calibration	

(2) The prefix, AIM ID, barcode type name, suffix, CRC check, and terminator are optional, and can be set by scanning or sending serial commands.

(3) Setting Remove User Raw Data allows you to remove the number of bytes specified for the raw data as set.

(4) Set up filter rules to filter specific characters or strings in user data.

(5) Set the original data output case, letters can be set to all uppercase output or lowercase output.

7.1 Remove raw data settings

This function removes the first N bytes and the last M bytes of the user's original data. When this function is successfully set, the first N bytes and the last M bytes of the user data will be eliminated. If N+M is greater than or equal to the length of the user data, there will be a beep but no data output. 0 for N means not to remove the first data of the user, 0 for M means not to remove the last data of the user. 20 for M and N, i.e., maximum 20 bytes in front of and 20 bytes in back of the user data will be removed.

7.1.1 Example 1 Removing the first 20 bytes of user data

$203020-F7AD (Example 1)

Example 1 Setting code description:

$2030 is the command identifier to set the function of removing the first N bytes from the user's raw data, and is fixed.

20 is the length of the byte (in decimal representation) before the raw user data is removed (0-20 bytes), 2

0 bytes

- Concatenator,

fixed F7AD

CRC calibration

7.1.2 Cancel removal of user front data



$203000-99CD (eliminates removal of front-end user data)

7.1.3 Example 2 Removing 20 bytes of data after user data



$203120-8119 (Example 2)

Example 2 Description.

$2031 is the command identifier to set the function of removing the last M bytes from the user's original data, fixed.

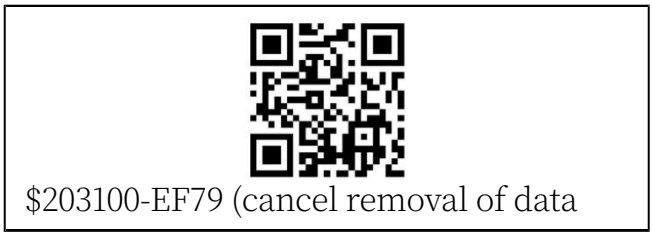
20 is the length of the byte (in decimal representation) following the removal of the original user data (0-20 bytes), 2

0 bytes

- Concatenation, fixed

8119CRC Checksum

7.1.4 Cancel removal of data behind the user



$203100-EF79 (cancel removal of data behind user)

7.2 Outputs only the settings of both ends of the data

This function is to provide the user with a setting to output only the first X or last Y bytes of the data, the range of X and Y is 0~255, after setting, the module will only output the first X or the last Y bytes of the data, if both the first and last functions are set, then the module will splice the first X bytes and the last Y bytes together and output them together, if the X and Y are 0, it means that the function is turned off, and if X+Y is 0, the default value is 0, if X+Y is 0, it means that the module will output the last Y bytes together. The default is 0, if X+, Y is 0, the module will output the first X bytes and the last Y bytes together.

Y is greater than the length of the data, no action will be taken when processing that data.
Raw data processing priority is: first ''remove the original data'', then do ''only output data ends'' processing, and finally do ''prefix" and "suffix", ''terminator'' and other data processing.
(Note: Supported by FW v4 series version with v4.x.30 and higher; supported by FW v6 series version.)

7.2.1 Example 1 Outputs only the first 20 bytes of data



$203214-8F51 (Example 1)

Example 1 Description.

$2032 Command identifier for the function that only outputs the first X bytes of data, unchanged.

14 is the length of the byte (in hexadecimal representation) after which the raw user data is removed, 20 bytes.

- Concatenation, fixed

8F51CRC Checksum

7.2.2 Eliminate data-only front-end output



$203200-74A5 (eliminates data-only front-end output)

7.2.3 Example 2 Outputs only the last 20 bytes of data



$203314-F9E5 (Example 2)

Example 1 Description.

$2033 Command identifier for the function of outputting only the last Y bytes of data, unchanged.

14 is the length of the byte (in hexadecimal representation) after which the raw user data is removed, 20 bytes.

- Concatenation, fixed

F9E5CRC calibration

7.2.4 Eliminate data-only output back-end



$203300-0211 (eliminates data-only output back end)

7.3 Adding Prefixes to Data

7.3.1 Adding a Prefix Example 1 Example 2

This function is to add a prefix to the recognition result. When this function is turned on, if the recognition is successful, the corresponding prefix will be added in front of the data. Two examples are provided below so that users can follow the instructions to add their own prefixes.



$202004-HEAD-7591 (Example 1)
The function of Example 1 is to add the prefix HEAD.



$202020-HEAD0123456789abcdef-5B3E (Example 2)
Example 2 adds the prefix HEAD0123456789abcdef.

Example 1 is used as an illustration:
Setting of string prefix (adding string prefix HEAD):
$2020 Prefixed logo, fixed
04 Byte length of prefix data (1-20 bytes), 4 bytes
- Connector, fixed
HEAD Prefix data, the length must be consistent with the above "prefix data byte length".
- Connector, fixed
7591 CRC checksum, $202004-HEAD- has a CRC16 checksum of 0x7591 (2 bytes), which can be converted to the string form 7591 (4 bytes); the CRC16 polynomial is 0x1021.

The final command is the string: $202004-HEAD-7591

Hexadecimal prefix setting (add hexadecimal prefix 0x48 0x45 0x41 0x44):
Convert all characters to hexadecimal, and the corresponding CRC check should also be the hexadecimal conversion of the
CRC checksum.

0x24 0x32 0x30 0x32 0x30 Identification of the prefix, fixed, i.e., the string $2020
0x30 0x34 The length in bytes (1-20 bytes) of the prefix data, i.e., string 04. 4 bytes
0x2d Concatenator, fixed
0x48 0x45 0x41 0x44 Prefix data, the length of which must be consistent with the "byte length of prefix data" above.
0x2d Concatenator, fixed
0x37 0x35 0x39 0x31 CRC checksum, hexadecimal data 0x24 0x32 0x30 0x32 0x30 0x30 0x34 0x2d0x48 0x45 0x41 0x44 0x2d has a CRC16 checksum of 0x7591 (2 bytes); converting this to the string form 7591 (4 bytes, if there are 'a', 'b', 'c', 'd', 'e', 'f', must be changed to 'a', 'b', 'c', 'd', 'e', 'f'. 'a', 'b', 'c', 'd', 'e', 'f', must be converted to uppercase), then convert the string form to hexadecimal form 0x 37 0x35 0x39 0x31; the CRC16 polynomial is 0x1021.

The final command is hexadecimal data: 0x24 0x32 0x30 0x32 0x30 0x30 0x34 0x2d 0x48 0x45 0x41 0x44 0x2d 0x37 0x35 0x39 0x31. The prefix HEAD can be set by sending the data in hexadecimal format through the interface.

7.3.2 Clear prefix



$202000-339C (removal of prefixes)

7.4 Adding Suffixes to Data

7.4.1 Adding a Suffix Example 1 Example 2



$202104-TAIL-FB3A (Example 1)
Add suffix TAIL



$202120-LIAT0123456789abcdef-E0BB (Example 2)
Add suffix LIAT0123456789abcdef

Example 1 illustrates:
$2021 Suffixed logos, fixed

04	Suffix data length byte length (1-20 bytes), 4 bytes
-	Connector, fixed

TAIL Suffix data, the length must be consistent with the "byte length of the suffix data" above.

- Connector, fixed

FB3ACRC calibration

For specific instructions, and how to set string and hexadecimal suffixes, refer to how to set prefixes.

7.4.2 remove suffix



$202100-4528 (remove suffix)

7.5 User Data Filtering Rule Setting

This function is used to filter and recognize the filter target character data contained in the user data. When this function is turned on, a string of data equal to the filtered character data in the user data will be eliminated if the reading is successful.

7.5.1 Example 1 Filtering a0 in the data



$202202-a0-B912 (Example 1)
Filtered data a0

Example 1 Description.

$2022 is the identifier of the command that sets the filtering function, fixed

02 is the byte length of the filtered data portion (1-20 bytes), 2 bytes

- Connector, fixed

a0 is the filtered data

- Connector, fixed

B912CRC calibration

For specific instructions, and how to set up string and hexadecimal filtering rules,

please refer to how to set up prefixes. Data to be filtered (example).
The following two examples are used to test the filter rule "a0", the scanning code or the serial send filter rule "$2022".

02-a0-B912", then scan the following two examples of data, which contains the string

"a0" will be

Filter out.

 1234aa0056 (data)	 a00b00c00 (data)

The result after setting "$202202-a0-B912":
"1234aa0056" The result after scanning is
"1234a056". "a00b00c00" The result is "0b00c00".

7.5.2 Clearing Filter Rules


$202200-DEF4 (clear filter)

7.6 Data CRC Checksum

This function is for adding CRC checksums (see Note (2) of the brief description) function to the end of the recognition data, including the prefix part, barcode type name, data part, and suffix part; but not including the terminator. The default is not output.

 $202301-9B71 (CRC checksum output)	 $202300-A840 (CRC test not output)

7.7 closing character

This feature is designed to allow the host to quickly distinguish the result of the current reading. When this function is enabled, if the reading is successful, the corresponding terminator will be added after the data. It is not added by default.

$201000-DD4E * Close the terminator	$201001-EE7F Add tab terminator TAB(0x0B)
$201002-BB2C Add carriage return terminator CR(0x0D)	$201003-881D Add carriage return line feed terminator CRLF(0x0D,0x0A)

7.8 Barcode Type Character Output

$20240 0-F96D* no output	$202401-CA5C Outputs

If set to output, the first byte of each recognized barcode identifies the type of the barcode, and the specific type of the barcode can be known by checking Appendix 1 Barcode Type Output Character Table. The default is no output.

7.9 Case output settings

This function allows you to set whether the letters in the barcode's data are output in all uppercase or lowercase. The default data is not processed for output as is.

 $204000-FE19 *Data output as is	 $204001-CD28 Upper case output for all letters	 $204002-987B Lowercase output for all letters

7.10 Disable URL output

This function can be set to disable url output or not (if disable, there will be no response when scanning the url, no light, sound, and output response), set to disable url output then it will respond normally and output the url. (Note: FW v4 series version, only v4.x.39 and higher version support; FW v6 series version support this function.)

$204200-1371 No URL output	$204201-2040 *Output URL

7.11 AIM ID Setting

AIM is the abbreviation of Automatic Identification Manufacturers (Association of Automatic Identification Manufacturers), AIM ID is a kind of prefix to distinguish different types of barcode, that is, the prefix data is added in front of the barcode data. The prefix format is:] + AIM prefix + number, e.g. the AIM ID of GS1_128 is:]C1, see Appendix 4.

$20250 0-8FD9* no output	$202501-BCE8 Outputs

8 Barcode reading configuration

Generally use the default open code system to read, the default open code system is some common code system, the user generally do not need additional settings. Unless the customer needs the code system can not be solved, then in the individual barcode configuration to open the code system. The more code systems you open, the worse the decoding effect, so you can close some unused barcode types to improve the decoding effect.

8.1 All barcode configurations

Allow reading of all types: All supported barcodes will be enabled.
***Open Default Reading Type**: The default commonly used barcode will be opened.
Disable reading of all 1D codes: will disable reading of all supported 1D barcodes. **Allow to read all 1D codes**: will enable to read all supported 1D barcodes.

$280101-3FD8 (Allows reading of all types)	$280102-6A8B (*Turn on default reading type)
$280103-59BA (Prohibits reading of all one-dimensional codes)	$280104-C02D (Allows reading of all 1D codes)

One-dimensional code inversion mode setting

Enable to scan the inverted 1D code, default not enable.

(Note: Supported only by FW v5 series releases, v5.2.5 and higher and v5.1.6 and higher).

$306601-1653 Open reverse color	$306600-2562 *Close reverse color

8.2 Individual Barcode Configuration

Read barcode length setting instructions, set the barcode reading the longest and shortest help to scan a specific length of the code system, you can more accurately scan to want to scan the barcode. For example, if you set the maximum read length of this type of barcode to 200, the barcode will be scanned more accurately.
If both the longest and shortest reading lengths are set to 100 characters, the barcode data must be 100 characters to be read, so the barcode can be read accurately. If both the longest and shortest readable character length are set to 100, the data of this type of barcode must be 100 characters to be read, so that the barcode can be read accurately. To set the minimum and maximum read lengths, proceed as follows:

（1） Scan the type code to enter the set code for the longest or shortest read length;

（2） Scan the numeric codes in Appendix 2 to set the length, if the length is to be set to 102, scan the numeric code 1, numeric code 0, and numeric code 3 in order. if the setting is wrong in the middle of the process, scan the discarded code and start the setting again from the (1) step;

（3） The maximum length of scanning digital code can be set to 9999, the minimum can be set to 1. After the setting of step (2) and scanning the save code, the buzzer will sound 3 times to indicate the successful setting, and the setting will take effect immediately.

(Note: FW v6 and v1 series versions only support code enable settings, not more detailed settings, F W v4 version supports all code settings)

8.2.1 CODE 128

(1) disable

$290701-5DE1 *Allowed to read

$290700-6ED0 Read disabled.

(2) Read Length

Read code maximum and minimum length range of 1-80, the minimum default read length of 1, the maximum default read length of 80, the setting method see read length setting description (such as setting out of the range of the value of the module will respond to the setup successfully, but will not take effect).

$290720-00B0 Minimum Read Length

$290730-3780 Maximum Read to Read Length

(3) Enable Disable ISBT 128

8.2.2 GS1_128 Code

(1) disable


$291301-3D41 *Allowed to read


$291300-0E70 Read disable.

(2) Read Length

Read code maximum and minimum length range of 1-80, the minimum default read length of 1, the maximum default read length of 80, the setting method see read length setting description (such as setting out of the range of the value of the module will respond to the setup successfully, but will not take effect).


$291320-6010 Minimum Read Length


$291330-5720 Maximum Read Length

8.2.3 CODE39

(1) disable


$290501-B089 *Allowed to read


$290500-83B8 Prohibited reading

(2) Read Length

Read code maximum and minimum length range of 1-48, the minimum default read length of 1, the maximum default read length of 48, the setting method see read length setting description (such as setting out of the range of the value of the module will respond to the setup successfully, but will not take effect).


$290520-EDD8 Minimum


$290530-DAE8 Max Read

（3）Check Digit Configuration

Note that the check digit type is only suitable for use with MOD4

*0: No checksum character. Reads and transmits barcodes with or without checksum characters. (At this time, when scanning barcodes with check characters, check characters are transmitted together)

1: Enables checksum and transmits the check character. If the checksum passes, the checksum character is used as normal data Last 1

Bits are transmitted together, a checksum failure will not send the barcode data

2: Enable the check, but do not transmit the check character. If the check passes, the normal data except the last check character will be transmitted; if the check fails, no barcode data will be transmitted

		
$290540-5F78 * Unchecked characters	$290541-6C49 Enable Check, Transmit Check Word	$290542-391A Enables checksums, but does not transmit checks

（4）Start/Stop Character Transfer Configuration

The start/stop characters are located at the front and end of the barcode. Configurable to transmit or not transmit start/stop characters

*0: No start/stop characters are transmitted 1: Start/stop characters are transmitted

	
$290550-6848 *No transmission	$290551-5B79 Transmission

（5）Code 32 (PARAF) Code Enable Off

Code 32 Pharmaceutical is a form of the Code 39 barcode used in Italian pharmacies. This code system is also known as PARAF

	
$290570-0628 *closed	$290571-3519 Enable

(6) FULL ASCII Code39

If FULL ASCII Code39 decoding is enabled, certain characters in the barcode will be decoded as single characters. For example, $V will be decoded as ASCII character SYN, and /C will be decoded as ASCII character #, see Appendix 3, code39 full ascii table for details.



$290580-2A19 *closed



$290581-1928 Enable

8.2.4 Interleaved 2 of 5

(1) disable



$290A01-380D *Allowed to read



$290A00-0B3C Read Prohibited

(2) Read Length

Read code maximum and minimum length range of 2-80, the minimum default read length of 4, the maximum default read length of 80, the setting method see read length settings description (such as setting out of the range of the value of the module will respond to the setup successfully, but will not take effect).



$290A20-655C Minimum Read Length



$290A30-526C Maximum Read Length

(3) check character

*0: No checksum character. Reads and transmits barcodes with or without checksum characters. (At this time, when scanning barcodes with checksum characters, checksum characters are transmitted together)
1: Enable checksum and transmit the check character. If the check passes, the normal data except the last check character will be transmitted; if the check fails, no barcode data will be sent.
2: Enables checksum, but does not transmit the check character. If the checksum

passes, the checksum character is treated as normal data Last 1
Bits are transmitted together, a checksum failure will not send the barcode data

$290A40-D7FC *No checksum character	$290A41-E4CD Enable checksum and transmit check character	$290A42-B19E Enable checksum, do not transmit check character

8.2.5 Straight 2 of 5 industrial

(1) disable

$290C01-D565 Permission to Read	$290C00-E654* Prohibited reading

(2) Read Length

Read code maximum and minimum length range of 1-48, the minimum default read length of 4, the maximum default read length of 48, the setting method see read length setting description (such as setting out of the range of the value of the module will respond to the setup successfully, but will not take effect).

$290C20-8834 Minimum Read Length	$290C30-BF04 Maximum Read Length

8.2.6 Straight 2 of 5 IATA

(1) Enable Disable (Note: only supported by FW v4 series version)



$291D01-2E19 Permission to Read



$291D00-1D28 *Disabled reading

(2) Read Length

Read code maximum and minimum length range of 1-48, the minimum default read length of 4, the maximum default read length of 48, the setting method see read length setting description (such as setting out of the range of the value of the module will respond to the setup successfully, but will not take effect).



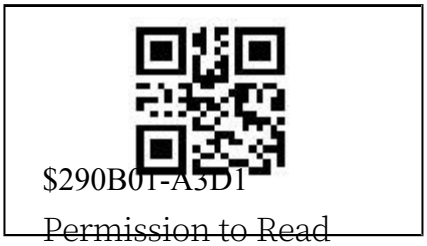
291D20-7348 Minimum Read Length



291D30-4478 Maximum Read Length

8.2.7 Matrix 2 of 5

(1) disable



$290B01-A3D1 Permission to Read



$290B00-90E0 *Disable reading

(2) Read Length

Read code maximum and minimum length range of 1-40, the minimum default read length of 1, the maximum default read length of 40, the setting method see read length settings description (such as setting out of the range of the value of the module will respond to the setup successfully, but will not take effect).



$290B20-FE80 Minimum Read Length



$290B30-C9B0 Maximum Read Length

(3) check character

*0: No checksum character. Reads and transmits barcodes with or without

checksum characters. (At this time, when scanning barcodes with
checksum characters, checksum characters are transmitted together)
1: Enables checksum and transmits the check character. If the checksum passes, the checksum character is used as normal data Last 1
Bits are transmitted together, a checksum failure will not send the barcode data
2: Enable the check, but do not transmit the check character. If the check
passes, the normal data except the last check character will be
transmitted; if the check fails, no barcode data will be transmitted

 $290B40-4C20 * Unchecked characters	 $290B41-7F11 Enable checksum, ~~transmit check~~ character	 $290A42-B19E Enables checksums, but ~~does not transmit check~~ words

8.2.8 Grid Matrix

(1) Enable Disable (Note: only supported by FW v4 series version)

 $293201-0F76 Permission to read	 $293200-3C47 *Disabled reading

(2) Read Length

Read code maximum and minimum length range of 1-2751, the minimum default read length of 1, the maximum default read length of 2751, the setting method see read length setting description (such as setting out of the range of the value of the module will respond to the setup success, but will not take effect).

 $293220-5227 Minimum Read Length	 $293230-6517 Maximum Read Length

8.2.9 CODE 93

(1) disable

 $290601-2B55 Permission to Read	 $290600-1864 *Prohibited reading

(2) Read Length

Read code maximum and minimum length range of 1-80, the minimum default read length of 1, the maximum default read length of 80, the setting method see read length setting description (such as setting out of the range of the value of the module will respond to the setup successfully, but will not take effect).



$290620-7604 Minimum Read Length



$290630-4134 Maximum Read Length

8.2.10 CODE 11

(1) disable



$290801-890F Permission to Read



$290800-BA3E *Disable reading

(2) Read Length

Read code maximum and minimum length range of 1-80, the minimum default read length of 4, the maximum default read length of 80, the setting method see read length setting description (such as setting out of the range of the value of the module will respond to the setup successfully, but will not take effect).



$290820-D45E Minimum Read Length



$290830-E36E Maximum Read Length

(3) check character

*0: 2 check digits & transfer 1: 1 check digits & transfer 2: 2

check digits no transfer 3: 1
check digits no transfer 4: no check digits, all transfer

$290840-66FE *2 check digits & transfer	$290841-55CF 1 check digits & transfer	$290842-009C 2 check digits no transfer
$290843-33AD 1 check digits no transfer		$290844-AA3A no check digits,all transfer

8.2.11 CODABAR

(1) disable

$290901-FFBB *Allowed to read

$290900-CC8A Prohibited reading

(2) Read Length

Read code maximum and minimum length range of 2-60, the minimum default read length of 4, the maximum default read length of 60, the setting method see read length setting description (such as setting out of the range of the value of the module will respond to the setup successfully, but will not take effect).

$290920-A2EA Minimum Read Length

$290930-95DA Maximum Read Length

(4) Enable disable transmission of start/stop characters

The start/stop characters are located at the front and end of the barcode. Configurable to transmit or not transmit start/stop characters

*0: No start/stop characters are transmitted 1: Start/stop characters are transmitted

$290950-277A * No

$290951-144B Transmission

(5) cascade

When cascading is enabled, the bar code scanner will look for a Codabar bar code with a "D" start character that is adjacent to a bar code with a "D" stop character.
In this case, the two barcodes are joined into one, omitting the "D" character.
CODABAR cascade switch, *0: off, 1: on


$290970-491A * Closed


$290971-7A2B Open

8.2.12 TELEPEN

(1) Enable Disable (Note: only supported by FW v4 series version)


$292A01-7C8E Read Permitted


$292A00-4FBF *Disable readability

(2) Read Length

Read code maximum and minimum length range of 2-80, the minimum default read length of 4, the maximum default read length of 80, the setting method see read length settings description (such as setting out of the range of the value of the module will respond to the setup successfully, but will not take effect).


$292A20-21DF Minimum Read Length


$292A30-16EF Maximum Read Length

8.2.13 MSI

(1) disable



$291101-D029 Permission to Read



$291100-E318 *Disable reading

(2) Read Length

Read code maximum and minimum length range of 4-48, the minimum default read length of 4, the maximum default read length of 48, the setting method see read length setting description (such as setting out of the range of the value of the module will respond to the setup success, but will not take effect).



$291120-8D78 Minimum Read Length



$291130-BA48 Maximum Read Length

8.2.14 GS1 DataBar (RSS)

(1) GS1 DataBar Omnidirectional (RSS_14) enable disable



$291701-F7B0 *Allowed to read



$291700-C481 Prohibition of readability

(2) GS1 DataBar Limited enable disable



$291703-91D2 *Allowed to read



$291702-A2E3 Prohibit reading

(3) GS1 DataBar Expanded enable disable



$291705-3B74 *Allowed to read



$291704-0845 Prohibited reading

（4）GS1 DataBar Expanded Read Length

Read code maximum and minimum length range of 4-74, the minimum default read length of 4, the maximum default read length of 74, the setting method see read length setting description (such as setting out of the range of the value of the module will respond to the setup successfully, but will not take effect).



$291720-AAE1 Minimum Read Length



$291730-9DD1 Maximum Read Length

8.2.15 PDF417

(1) disable



$292501-F40A Permission to Read



$292500-C73B *Disable reading

(2) Read Length

Read code maximum and minimum length range of 1-2750, the minimum default read length of 1, the maximum default read length of 2740, the setting method see read length setting description (such as setting out of the range of the value of the module will respond to the setup success, but will not take effect).



$292520-A95B Minimum Read Length



$292530-9E6B Maximum Read Length

8.2.16 MICRO PDF

(1) Enable Disable (Note: only supported by FW v4 series version)



$291401-6C6C Permission to Read



$291400-5F5D *Disable reading

(2) Read Length

Read the code maximum and minimum length range of 1-250, the minimum default read length of 1, the maximum default read length of 250, the setting method see the read length settings description (such as setting values beyond the range of the module will respond to the setup success, but will not take effect).



$291420-313D Minimum Read Length



$291430-060D Maximum Read Length

8.2.17 Aztec Code

(1) Enable Disable (Note: only supported by FW v4 series version)



$293601-C587 Permission to Read



$293600-F6B6 *Disable Readers

(2) Read Length

Read the code maximum and minimum length range of 1-3832, the minimum default read length of 1, the maximum default read length of 3832, set the method see read length settings description (such as setting the value beyond the range of the module will respond to the setup is successful, but will not take effect).



$293620-98D6 Minimum Read Length



$293630-AFE6 Maximum Read Length

8.2.18 MaxiCode

(1) Enable Disable (Note: only supported by FW v4 series version)



$291201-4BF5 Permission to Read



$291200-78C4 *Disable reading

(2) Read Length

Read code maximum and minimum length range of 1-93, the minimum default read length of 4, the maximum default read length of 93, the setting method see read length setting description (such as setting out of the range of the value of the module will respond to the setup successfully, but will not take effect).



$291220-16A4 Minimum Read Length



$291230-2194 Maximum Read Length

8.2.19 QR

(1) QR Enable Off

To prevent the device from locking up, CODE128 will be turned on when QR code is disabled and QR code will be turned on when CODE128 is disabled.



$292600-5CE7 Prohibited QR



$292601-6FD6 * Allowed to read QR



$292601-6FD6 * Allowed to read QR

(2) MICRO QR Enable Off



$292671-EA46 *Allowed to read



$292670-D977 Prohibition of readability

(3) QR Reverse Color Function

Enable to scan the inverted QR code, default enable.



$292681-C677 *Turn on reverse color



$292680-F546 Turn off reverse color

8.2.20 Han Xin code

(1) Enable Disable (Note: only supported by FW v4 series version)



$293101-94AA Permission to Read



$293100-A79B *Disable Readers

(2) Read Length

Read code maximum and minimum length range of 1-1000, the minimum default read length of 1, the maximum default read length of 1000, the setting method see read length setting description (such as setting out of the range of the value of the module will respond to the setup successfully, but will not take effect).



$293120-C9FB Minimum Read Setting



$293130-FECB Maximum Read Setting

8.2.21 China Post (Hong Kong 2 of 5)

(1) Enable Disable (Note: only supported by FW v4 series version)



$292801-CD8C Allowed to read



$292800-FEBD *Disable Readers

(2) Read Length

Read code maximum and minimum length range of 2-80, the minimum default read length of 4, the maximum default read length of 80, the setting method see read length settings description (such as setting out of the range of the value of the module will respond to the setup successfully, but will not take effect).



$292820-90DD Minimum

Read Setting



$292830-A7ED Maximum

Read Setting

8.2.22 Data Matrix

(1) disable


$292701-1962 Permission to read


$292700-2A53 *Disable reading

(2) Read Length

Read the code maximum and minimum length range of 1-3116, the minimum default read length of 1, the maximum default read length of 3116, the setting method see read length setting description (such as setting out of the range of the value of the module will respond to the setup success, but will not take effect).


$292720-4433 Minimum Read Settings


$292730-7303 Maximum read setting

8.2.23 NEC 2 of 5

(1) Enable Disable (Note: only supported by FW v4 series version)


$293801-67DD *Allowed to read


$293800-54EC Prohibited reading

(2) Read Length

Read code maximum and minimum length range of 2-80, the minimum default read length of 4, the maximum default read length of 80, the setting method see read length settings description (such as setting out of the range of the value of the module will respond to the setup successfully, but will not take effect).



$293820-3A8C Minimum Read Setting



293830-0DBC Maximum Read Setting

(3) check character

*0: No checksum character. Reads and transmits barcodes with or without checksum characters. (At this time, when scanning barcodes with check characters, check characters are transmitted together)
1: Enables checksum and transmits the check character. If the checksum passes, the checksum character is used as normal data Last 1
Bits are transmitted together, a checksum failure will not send the barcode data
2: Enable the check, but do not transmit the check character. If the check passes, the normal data except the last check character will be transmitted; if the check fails, no barcode data will be transmitted

		
$293840-882C *No checksum character	$293841-BB1D Enable checksum, transmit check character	$293842-EE4E Enables checksums, but does not transmit check characters

8.2.24 UPC-A

(1) disable

 $290101-7A78 *Allowed to read	 $290100-4949 Prohibition of readability

NOTE: When scanning UPC-A off, the UPC-A barcode is transmitted as EAN-13.

(2) Whether the check digit is transmitted

 $290141-A6B8 *Transmission	 $290140-9589 No transmission

（3）digital system

Digital System Character Transfer Switch

$290171-FFE8 *Transmission

$290170-CCD9 No transmission

（4）additional code

2-digit add-on code switch, *0: OFF, 1: ON

$290160-FBE9 *off

$290161-C8D8 Open

5-digit add-on code switch, *0: OFF, 1: ON

$290162-9D8B *off

$290163-AEBA On

Required add-on code (UPC-A with add-on code only) switch, *0: off, 1: on

$290164-372D *off

$290165-041C Open

Additional code separator (whether to turn on the space separator additional code) switch, 0: off, *1: on

$290166-514F Off

$290167-627E *open

8.2.25 UPC-E0 & UPC-E1

(1) disable

$290201-E1A4
*Allows reading of UPC-E0

$290200-D295
Prohibit reading of UPC-E0

$291801-235E
Allows reading of UPC-E1

$291800-106F
* Prohibit reading of UPC-E1

(2) Whether the check digit is transmitted or not

$290241-3D64
*Transmission

$290240-0E55 No transmission

(3) additional code

2-digit add-on code switch, *0: OFF, 1: ON

$290260-6035 *off

$290261-5304 Open

5-digit add-on code switch, *0: OFF, 1: ON

$290262-0657 *Off

$290263-3566 Open

Required add-on code (only solves UPC-E with add-on code) switch, *0: off, 1: on


$290264-ACF1 *off


$290265-9FC0 On

Additional code separator (whether to turn on the space separator additional code) switch, 0: off, *1: on


$290266-CA93 Off


$290267-F9A2 *open

（4）digital system

Digital System Character Transfer Switch


$290271-6434 *Transmission


$290270-5705 No transmission

（5）Barcode expansion to 12-bit UPC-A

UPC-E barcode expanded to 12-bit UPC-A format switch, *0: off, 1: on


$290280-7B34 *Non-transmission


$290281-4805 Transmission

8.2.26 EAN/JAN-13

（1）disable

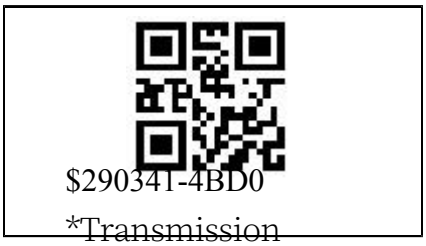
NOTE: To convert UPC-A barcodes to EAN-13 format, scan the UPC-A off barcode!





（2）check character

Checks whether a character is transmitted or not





（3）additional code

2-digit add-on code switch, *0: OFF, 1: ON





5-digit add-on code switch, *0: OFF, 1: ON





Required add-on code (EAN/JAN-13 with add-on code only) switch, *0: off, 1: on





Additional code separator (whether to turn on space separator additional code) switch, 0: off, *1: on



$290366-BC27 Off



$290367-8F16 *Open

（4）ISBN Translate

EAN-13 Bookland symbols will be converted to the equivalent ISBN symbol format when set to ISBN symbol transfer



$290370-21B1 *Non-converting



$290371-1280 Conversion

8.2.27 EAN/JAN-8

（1）disable



$290401-C63D *Allowed to read



$290400-F50C Prohibited reading

（2）Checks whether a character is transmitted or not



$290441-1AFD *Transmission



$290440-29CC No transmission

（3）additional code

2-digit add-on code switch, *0: OFF, 1: ON



$290460-47AC *off



$290461-749D On

5-digit add-on code switch, *0: OFF, 1: ON


$290462-21CE *Off


$290463-12FF On

Required add-on code (EAN/JAN-8 with add-on code only) switch, *0: off, 1: on


$290464-8B68 *off


$290465-B859 Open

Additional code separator (whether to turn on the space separator additional code) switch, 0: off, *1: on


$290466-ED0A Off


$290467-DE3B *open

Appendix 1 Barcode Type Output Character Table

If the bar code type character is set to Output, the first byte of each recognized bar code indicates the type of that bar code. The barcode type corresponding to the character is shown in the following table:

Barcode Type	output character
AZTEC_CODE	B
China Post	F
CODABAR	G
CODE11	I
CODE39	M
CODE39_BASE32 (CODE 32)	L
CODE93	O
CODE128	J
DATAMATRIX	S
EAN8	W
EAN13	V
GRID MATRIX	X
GS1 128 (EAN-128)	Y
HANXIN (Hanshin code)	Z
Interleaved 2 of 5	c
Matrix 2 of 5	h
MAXICODE	i
MICROPDF	j
MSI-Plessey	k
PDF417	m
QRCODE/ MICRO QR	q
RSS -14 (GS1_DATABAR)	r
RSS_Expand (GS1_DATABAR_EXP)	>
RSS_Limited (GS1_DATABAR_LIM)	<
Standard 2 of 5/ Industrial 2 of 5/ IATA 2 of 5	s
TELEPEN	t
TRIOPTIC	u
UPCA	w
UPCE	x
NEC 2 of 5	l

Appendix 2 Digital Codes

 $282001-0DEF (Digit code 1)		 $282002-58BC (numeric code 2)
 $282003-6B8D (Digit code 3)		 $282004-F21A (Digit code 4)
 $282005-C12B (Digit code 5)		 $282006-9478 (Digit code 6)
 $282007-A749 (digit code 7)		 $282008-B777 (Digit code 8)
 $282009-8546 (Digit code 9)		 $28200A-BEDE (digit code 0)
 $28200B-A3BE (Saved)		 $28200C-948F (Waiver)

Appendix 3 CODE 39 FULL ASCII Correspondence Table

ASCII	ENCODING	ASCII	ENCODING	ASCII	ENCODING	ASCII	ENCODING
NUL	%U	SP	Space	@	%V	`	%W
SOH	$A	!	/A	A	A	a	+A
STX	$B	"	/B	B	B	b	+B
ETX	$C	#	/C	C	C	c	+C
EOT	$D	$	/D	D	D	d	+D
ENQ	$E	%	/E	E	E	e	+E
ACK	$F	&	/F	F	F	f	+F
BEL	$G	'	/G	G	G	g	+G
BS	$H	(	/H	H	H	H	H
HT	$I	)	/I	I	I	i	+I
LF	$J	*	/J	J	J	j	+J
VT	$K	+	/K	K	K	k	+K
FF	$L	,	/L	L	L	l	+L
CR	$M	-	-	M	M	m	+M
SO	$N	.	.	N	N	n	+N
SI	$O	/	/O	O	O	o	+O
DLE	$P	0	0	P	P	p	+P
DC1	$Q	1	1	Q	Q	q	+Q
DC2	$R	2	2	R	R	r	+R
DC3	$S	3	3	S	S	s	+S
DC4	$T	4	4	T	T	t	+T
NAK	$U	5	5	U	U	u	+U
SYN	$V	6	6	V	V	v	+V
ETB	$W	7	7	W	W	w	+W
CAN	$X	8	8	X	X	x	+X
EM	$Y	9	9	Y	Y	y	+Y
SUB	$Z	:	/Z	Z	Z	z	+Z
ESC	%A	;	%F	[	%K	{	%P
FS	%B	<<	%G	\windshield	%L	|	%Q
GS	%C	=	%H	]	%M	}	%R
RS	%D	>	%I	^	%N	~	%S
YS	%E	?	%J	_	%O	DEL	%T, %X. %Y, %Z

Appendix 4 AIM ID List

Barcode Type	AIM ID	note
Codabar	]Fm	0-1
Code 11	]Hm	0,1,3
Code 128	]Cm	0,1,2,4
ISBT 128	]C4	
Code 39	]Am	0,1,3,4,5,7
Code 93/93i	]Gm	0-9, A-Z, a-m
EAN		
EAN-8 (including SBN)	]E0	
EAN-13 with Add-On	]E3	
EAN-13 with Composite	]E3	
EAN-8	]E4	
EAN-8 with Add-On	]E3	
GS1		
GS1 DataBar	]e0	
GS1 DataBar Limited	]e0	
GS1 DataBar Expanded	]e0	
GS1-128	]C1	
GS1 Composite	]em	0-3
GS1 DataBar Omnidirectional	]e0	
2 of 5		
China Post	]X0	
Interleaved 2 of 5	]Im	0,1,3
Matrix 2 of 5	]X0	
NEC 2 of 5	]X0	
Straight 2 of 5 IATA	]Rm	0,1,3
Straight 2 of 5 Industrial	]S0	
MSI	]Mm	0,1
Telepen	]B0	
UPC		
UPC-A	]E0	
UPC-A with Add-On	]E3	
UPC-A with Composite	]E3	
UPC-E	]E0	
UPC-E with Add-On	]E3	
UPC-E1	]X0	
Aztec Code	]zm	0-9, A-C

Han Xin	]X0	
Data Matrix	]dm	0-6
MaxiCode	]Um	0-3
PDF417	]Lm	0-2
Micro PDF417	]Lm	3,4,5
QR	]Qm	0-6
Micro QR	]Qm	
Grid Matrix	]X0	

Reference: ISO/IEC 15424:2008 Information technology - Automatic identification and data capture techniques - Data carrier identifiers (including symbolic identifiers)

Appendix 5 POS Interface Examples

Take the 82-byte QR code described in the setup as an example, the content of the QR code is: 1234567890abcedfgAB CEFG1234567890abcedfgABCEFG1234567890abcedfgABCEFG1234567890abc, and the content of the QR code is: 1234567890abcedfgAB CEFG1234567890abc.

When the module is in the POS interface, the module outputs the following results, which are analyzed as follows:

Packet header (32 bytes): 56 00 00 00

00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00

The end of the packet is fixed to a packet whose first bit is 0x0D and the other 31 bytes are 00.

Appendix 6 Test Barcodes

Matrix 2 of 5  1234567890	CODE 128  614141999996
CODE 39  1234567890	GS1_128  614141999996
Industrial 2 of 5  1234567890	Interleaved 2 of 5  1234567890
standard 2 of 5(IATA)  1234567890	CODE 11  123456789019

Tel epen



1234567890

PDF417



This is PDF417!

Grid Matrix



this is grid matrix!

1234567890

CODE 93



1234567890

Aztec Code



This This Aztec Code!

GS1 DataBar Expanded



(12) 345678 (90) 123



codabar

1234567890

GS1 DataBar Limited



(01) 1 2345678 90123 1

MSI



1234567890

GS1 DataBar (RSS)



(f01) 1 2345678 90123 1

Data Matr i x



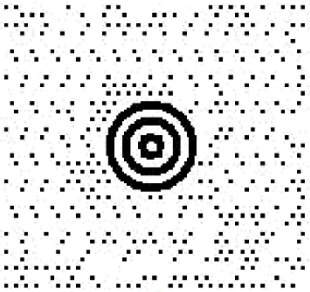
Th i s　i s Data Matr í x !

Han X in Oode



Th i s　i s Han X i n Code!

MaxiCode(标准符号：4)



Th i s　Th i s Max i Code!

UPC A



E@/JAN-13



NEC 2 of 5

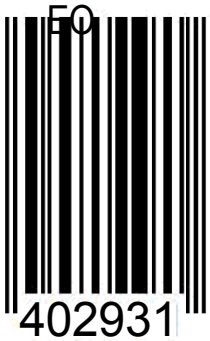
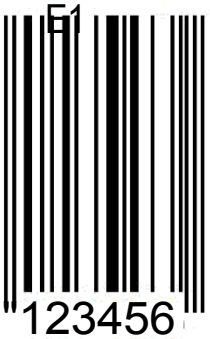


EAN/JAN-8



Chi na Post



OR Code  This is QR Code!	H micro PDF417  This is Micro PDF417!
UPC-E0  0 402931 1	UPC-E1  123456 2